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DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

"Analyzing Public Transport Reliability Using Power BI"

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **"Analyzing Public Transport Reliability Using Power BI"** has been successfully presented by **S SABA ANJUM** bearing USN **3BR24CD403** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

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1: Introduction

1.1 About the Project Public transport plays a crucial role in connecting people within cities, but its reliability is often challenged by delays caused by traffic congestion, mechanical breakdowns, weather conditions, and staffing issues. These delays directly affect passenger satisfaction and trust in the system.

The purpose of this project is to analyze the reliability of buses, metros, and trams using a structured dataset in CSV format. The dataset includes trip details such as scheduled departure, actual arrival, vehicle type, and delay causes. By importing this data into Power BI, the project applies data cleaning, transformation, and visualization techniques to uncover meaningful insights.

The final outcome is an interactive dashboard that highlights on-time performance, average delay minutes, and the distribution of delay causes. This dashboard not only presents the current state of transport reliability but also helps identify problem areas and provides recommendations for improvement. The project demonstrates how data visualization can transform raw transport data into actionable insights for planners and decision-makers.

1.2 Language Used The project was implemented entirely using Microsoft Power BI.

- CSV Dataset: The raw transport data was provided in CSV format, which was directly imported into Power BI.
- Power Query: Used for data cleaning and transformation, including handling missing values, renaming fields, and calculating delay minutes.
- DAX (Data Analysis Expressions): Applied to create key measures such as On-Time Performance %, Average Delay Minutes, and Trip Counts.
- Power BI Visualization Tools: Used to design bar charts, line charts, pie charts, and KPI cards, along with filters and slicers for interactivity.

By combining CSV data with Power BI's powerful features, the project successfully delivered a clear, interactive dashboard that explains transport reliability in a simple and effective way.

2: Scope of the Project

2.1 Problem Statement Public transport systems are the backbone of urban mobility, providing affordable and accessible travel for millions of people.

However, their effectiveness is often reduced due to frequent delays. These delays are caused by multiple factors such as traffic congestion, mechanical breakdowns, weather conditions, and staffing shortages. Passengers rely on buses, metros, and trams to reach destinations on time, and when services fail to meet expectations, it leads to dissatisfaction, loss of productivity, and reduced trust in the system.

Traditional reporting methods often fail to capture the full picture of reliability because they lack interactive analysis and visualization. Without a clear dashboard, authorities cannot easily identify which vehicle type faces the most delays, which days are most problematic, or what causes are most frequent. This gap makes it difficult to design effective solutions. Hence, there is a strong need for a structured, data-driven approach that can transform raw transport data into meaningful insights and guide decision-makers toward improving reliability.

2.2 Objectives The objectives of this project are designed to address the problem in a systematic way:

- Data Import and Cleaning: Load the CSV dataset into Power BI and prepare it using Power Query by removing errors, handling missing values, and calculating delay minutes.
- Performance Measurement: Compute KPIs such as On-Time Performance %, Average Delay Minutes, and Trip Counts using DAX measures.
- Visualization of Insights: Build charts and KPI cards to show performance by vehicle type, delay causes, and weekly patterns..

3: System Requirements

3.1 Functional Requirements The functional requirements define what the system must be able to do in order to achieve the project objectives.

- Import CSV dataset into Power BI for analysis.
- Clean and transform data using Power Query (handling missing values, renaming fields, calculating delay minutes).
- Create DAX measures to calculate KPIs such as On-Time Performance %, Average Delay Minutes, and Trip Counts.
- Build interactive visuals including KPI cards, bar charts, line charts, and pie charts.

- Provide filters and slicers to allow users to analyze data by vehicle type, day of week, and delay cause.
- Generate a final dashboard that presents insights in a clear and user-friendly manner.

3.2 Non-Functional Requirements Non-functional requirements ensure the system is efficient, reliable, and easy to use.

- **Usability**: Dashboard must be simple and intuitive for non-technical users.
- **Performance**: Visualizations should load quickly even with large datasets.
- **Scalability**: System should handle additional datasets if more transport data is added in the future.
- **Accuracy**: Calculations of KPIs and delay minutes must be precise.

3.3 Hardware Requirements To run Power BI effectively and handle CSV datasets, the following hardware is recommended:

- A computer or laptop with at least **8 GB RAM**.
- Processor: Intel i5 or equivalent (minimum dual-core).
- Storage: At least 500 GB HDD/SSD for dataset storage and software installation.
- Display: Full HD resolution (1920x1080) for clear visualization of dashboards.
- Stable internet connection for updates and sharing reports.

3.4 Software Requirements The software environment required for the project includes:

- **Microsoft Power BI Desktop** (for data import, transformation, and visualization).
- **Power Query** (integrated within Power BI for data cleaning and preparation).
- **DAX (Data Analysis Expressions)** (for creating measures and calculations).

- **CSV Dataset File** (raw transport data containing trip details, vehicle type, delay cause, and timings).
- Optional: **Power BI Service** (for publishing and sharing dashboards online).

4: Methodology

4.1 Data Collection The foundation of this project was a CSV dataset containing detailed records of public transport trips. Each record included attributes such as trip ID, vehicle type (bus, metro, tram), scheduled departure time, actual arrival time, and the cause of delay. This dataset was chosen because it provided structured information that could be directly imported into Power BI without the need for complex database connections.

The dataset was representative of real-world transport operations, capturing both punctual trips and delayed ones. By focusing on these attributes, the project was able to measure reliability in terms of on-time performance and average delay minutes. .

4.2 Data Preparation (Power Query) Once the CSV file was imported into Power BI, the next step was data preparation using Power Query. This stage involved cleaning and transforming the dataset to make it suitable for analysis. Missing values were identified and handled appropriately, while irrelevant or duplicate records were removed to maintain accuracy.

New calculated columns were created, such as “Delay Minutes,” which was derived by subtracting scheduled departure time from actual arrival time. This transformation was critical because it allowed the project to quantify delays in a consistent manner. Power Query also standardized column names and ensured that data types were correctly assigned (e.g., time fields as Date/Time, vehicle type as Text).

By the end of this stage, the dataset was clean, structured, and ready for visualization.

4.3 Data Visualization Visualization was the core of the methodology, as the project aimed to transform raw data into meaningful insights. Power BI’s visualization tools were used to create a variety of charts and KPI cards.

- **KPI Cards** displayed overall on-time performance percentage and average delay minutes.

- **Bar Charts** compared performance across vehicle types, showing which mode of transport was most reliable.
- **Line Charts** illustrated delay trends across days of the week, highlighting mid-week peaks.
- **Pie Charts** showed the distribution of trips by vehicle type, emphasizing the dominance of buses in the dataset.
- **Cause Analysis Charts** explained why delays occurred, such as traffic congestion or mechanical breakdowns.

These visuals made the data easy to interpret and provided a clear narrative about transport reliability.

4.4 Filters and Interactivity To enhance usability, filters and slicers were added to the dashboard. These allowed users to interact with the data and focus on specific aspects. For example, a slicer for vehicle type enabled users to view performance metrics for buses, metros, or trams individually. Similarly, filters for day of the week allowed analysis of daily patterns, while delay cause filters helped isolate specific issues.

This interactivity transformed the dashboard from a static report into a dynamic analytical tool. Decision-makers could explore the dataset from multiple angles, uncovering insights that were not immediately visible in aggregated results.

4.5 Dashboard Design The dashboard was designed with clarity and storytelling in mind. The layout followed a logical sequence: KPI cards at the top to provide a quick overview, followed by charts that explained performance in detail. Delay causes and weekly patterns were placed centrally to highlight problem areas, while trip distribution visuals were positioned to show the scale of operations.

Color coding was used to differentiate vehicle types and delay categories, making the visuals more intuitive. The design emphasized simplicity, ensuring that even non-technical users could understand the insights at a glance.

4.6 Outcome The final outcome was a fully interactive Power BI dashboard that provided actionable insights into public transport reliability. The analysis revealed that buses were most affected by traffic delays, metros suffered from mechanical breakdowns, and mid-week delays were consistently higher. Overall

on-time performance was found to be below expectations, indicating a need for improvement.

5: Implementation

5.1 Dataset Description The dataset used for this project was provided in CSV format. It contained records of public transport trips with attributes such as trip ID, vehicle type (bus, metro, tram), scheduled departure time, actual arrival time, and delay cause. These fields were essential for calculating delay minutes and measuring on-time performance. The dataset was structured, easy to import into Power BI, and representative of real-world transport operations.

5.2 Data Loading The CSV file was imported directly into Power BI Desktop. Power BI's built-in connectors allowed seamless loading of the dataset without requiring external databases. Once loaded, the dataset appeared in the Power BI data model, ready for cleaning and transformation. This step ensured that the raw data was accessible for further processing.

5.3 Data Transformation (Power Query) Power Query was used to clean and transform the dataset. Missing values were handled, duplicate records were removed, and irrelevant columns were excluded. A calculated column "Delay Minutes" was created by subtracting scheduled departure time from actual arrival time. This transformation standardized the dataset and ensured accuracy in subsequent analysis. Power Query also allowed renaming fields and assigning correct data types (e.g., Date/Time for timestamps, Text for vehicle type).

5.4 Data Modelling After transformation, the dataset was structured into a logical data model. Relationships between tables were defined where necessary, although the project primarily relied on a single CSV file. The data model ensured that measures and calculations could be applied consistently across the dataset. This stage prepared the foundation for DAX calculations and dashboard visuals.

5.5 DAX Measures (Key Calculations) DAX (Data Analysis Expressions) was used to create key measures that drove the dashboard insights:

- **On-Time Performance %**: Percentage of trips that arrived on or before scheduled time.
- **Average Delay Minutes**: Mean delay duration across all trips.

- **Trip Counts**: Total number of trips by vehicle type and day of week.
- **Delay Cause Distribution**: Count of delays categorized by traffic, breakdowns, or staffing issues.

These measures provided the quantitative backbone of the dashboard.

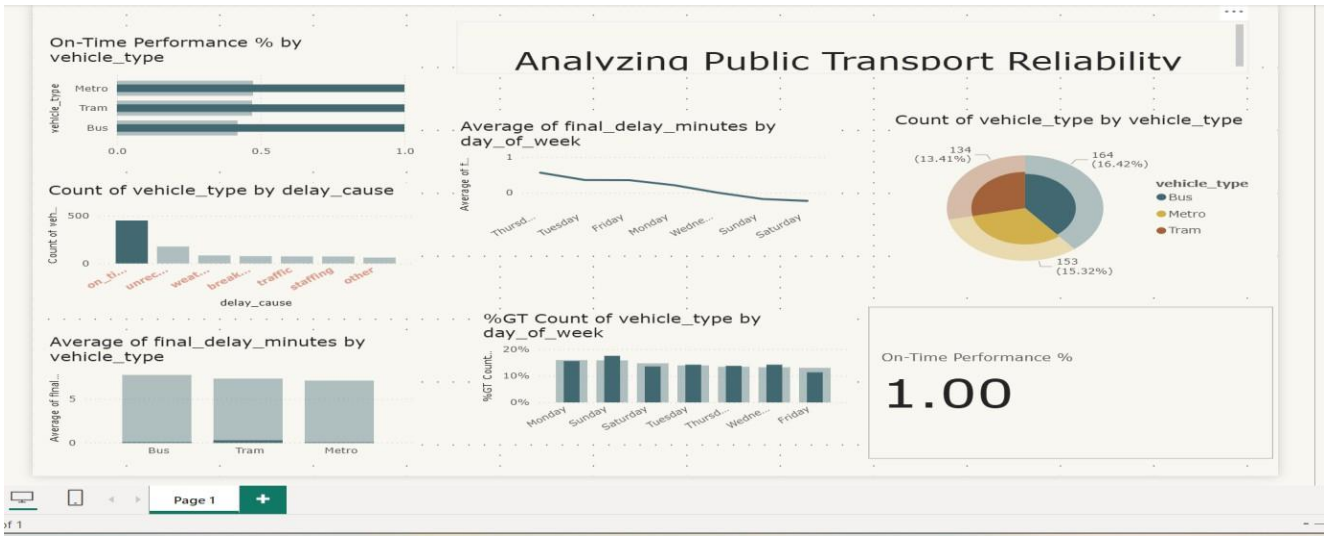
5.6 Dashboard Visualization Power BI's visualization tools were used to create charts and KPI cards. KPI cards displayed overall performance metrics, bar charts compared vehicle types, line charts showed weekly delay trends, and pie charts illustrated trip distribution. Cause analysis charts explained why delays occurred. Together, these visuals transformed raw data into meaningful insights.

5.7 Filters / Slicers Interactive slicers were added to the dashboard to allow users to filter data by vehicle type, day of week, and delay cause. These filters enabled deeper analysis, allowing decision-makers to isolate specific problem areas and explore data from multiple perspectives. Interactivity was a key feature that made the dashboard dynamic and user-friendly.

5.8 Dashboard Layout The dashboard was designed with clarity and storytelling in mind. KPI cards were placed at the top for quick overview, followed by charts that explained performance in detail. Delay causes and weekly patterns were positioned centrally to highlight problem areas, while trip distribution visuals were placed to show the scale of operations. Color coding differentiated vehicle types and delay categories, making the visuals intuitive.

5.10 Final Output The final output was an interactive Power BI dashboard that provided a comprehensive view of public transport reliability. It revealed that buses were most affected by traffic delays, metros faced mechanical breakdowns, and mid-week delays were highest. Overall on-time performance was below expectations, highlighting areas for improvement. The dashboard not only measured reliability but also suggested practical solutions such as route optimization, preventive maintenance, and staffing adjustments

Result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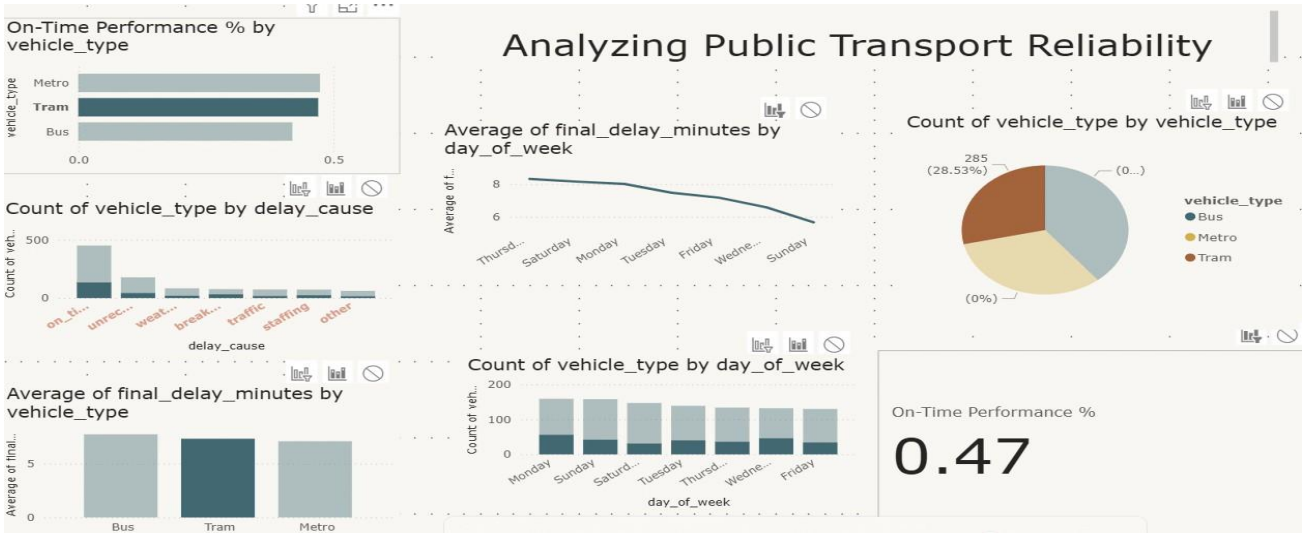
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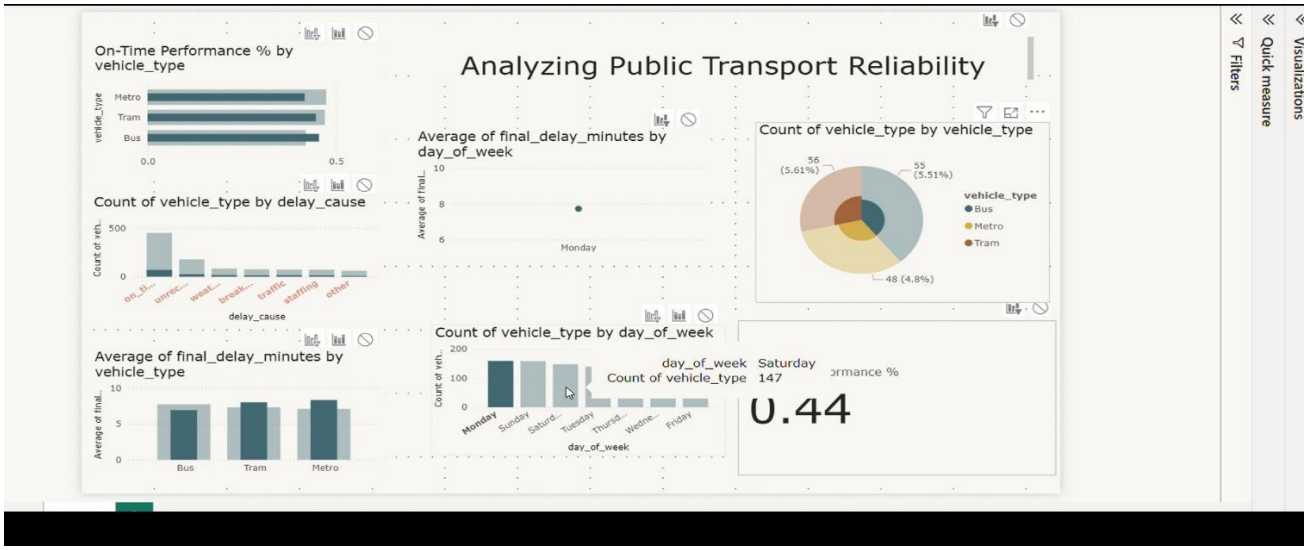
On-Time Performance % by vehicle_type
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Analyzing Public Transport Reliability
Average of final_delay_minutes by day_of_week
Count of vehicle_type by vehicle_type
vehicle_type
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Tram
Count of vehicle_type by day_of_week
day_of_week Saturday
Count of vehicle_type 147
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Filters
Quick measure
Visualizations

Conclusion

The project successfully demonstrated how a simple CSV dataset, when combined with the powerful features of Microsoft Power BI, can provide deep insights into public transport reliability. By systematically importing, cleaning, and transforming the data using Power Query, and applying DAX measures for key calculations, the project was able to quantify on-time performance, average delay minutes, and identify the most common causes of delays.

The interactive dashboard created in Power BI served as the central outcome of this work. It allowed users to explore transport performance across different vehicle types, days of the week, and delay causes. The visual storytelling approach made complex data easy to understand, even for non-technical stakeholders. The findings revealed that buses were most affected by traffic congestion, metros faced mechanical breakdowns, and mid-week delays were consistently higher.

Beyond identifying problems, the dashboard provided actionable recommendations. These included optimizing bus routes to reduce congestion, implementing preventive maintenance schedules for metros, and adjusting staffing levels during peak mid-week periods. Such recommendations highlight the practical value of the project, as they can directly support transit authorities in improving service reliability and passenger satisfaction.

In conclusion, this project illustrates the importance of data-driven decision-making in public transport management. It shows that even with a basic CSV dataset, meaningful insights can be extracted and presented through Power BI dashboards. The methodology and implementation steps outlined in this report can serve as a model for similar projects in other domains, where reliability and performance are critical.

References

1. Microsoft Power BI Documentation. *Getting Started with Power BI*. Available at: www.microsoft.com/powerbi *(Used for understanding dashboard creation, visualization tools, and DAX measures.)*
2. Power Query Documentation. *Data Transformation and Cleaning in Power BI*. Available at: www.docs.microsoft.com/power-query (docs.microsoft.com in Bing) *(Referenced for data preparation techniques such as handling missing values and creating calculated columns.)*
3. Transport for London (TfL). *Public Transport Reliability Reports*. Available at: www.tfl.gov.uk *(Used as a benchmark for understanding transport reliability metrics and delay causes.)*
4. IEEE Xplore Digital Library. *Data Analytics in Public Transport Systems*. Available at: www.ieee.org/xplore *(Consulted for academic insights into transport performance analysis and visualization approaches.)*
5. Kaggle Datasets. *Public Transport Delay Data (CSV)*. Available at: www.kaggle.com